

Day 8 — Malware Comparison Table

1. Virus

- What it is: A virus is malicious software that attaches itself to a legitimate file or program.
- How it works: It becomes active when the infected file or program is executed.
- How it spreads: It can spread when infected files are shared, downloaded, or transferred.
- Target: Files, programs, and computer systems.
- Impact: It can corrupt, modify, or delete files and reduce system performance.
- Example: A malicious executable attached to a document that infects the computer when opened.
- Prevention: Use antivirus software, keep software updated, avoid suspicious downloads, and scan files before opening them.

2. Worm

- What it is: A worm is malware that can replicate itself without attaching to another program.
- How it works: It can automatically search for vulnerable systems and spread across networks.
- How it spreads: Mainly through network vulnerabilities, infected devices, or insecure connections.
- Target: Computers, servers, and networks.
- Impact: It can consume network bandwidth, slow systems, and spread rapidly throughout an organization.
- Example: A worm exploiting an unpatched vulnerability and automatically spreading to other computers on the same network.
- Prevention: Keep operating systems and applications updated, use firewalls, and monitor network activity.

3. Trojan

- What it is: A Trojan is malware that disguises itself as legitimate or useful software.
- How it works: It tricks the user into downloading or executing the malicious program.
- How it spreads: Usually through malicious downloads, email attachments, fake applications, or compromised websites.
- Target: Computers, applications, accounts, and sensitive information.
- Impact: It can steal information, install additional malware, or give attackers unauthorized access.
- Example: A fake software application that appears legitimate but secretly installs malware.
- Prevention: Download software only from trusted sources, avoid suspicious attachments, and use security software.

4. Spyware

- What it is: Spyware is malware designed to secretly monitor a user's activities and collect information.
- How it works: It operates in the background without the user's knowledge.
- How it spreads: It can be installed through malicious software, suspicious downloads, compromised websites, or bundled applications.
- Target: Passwords, browsing activity, personal information, and other sensitive data.
- Impact: It can cause privacy violations, credential theft, and unauthorized collection of personal information.
- Example: Software secretly recording a user's browsing activity and sending the information to an attacker.
- Prevention: Use anti-malware software, keep systems updated, avoid suspicious downloads, and review installed applications.

5. Ransomware

- What it is: Ransomware is malware that prevents users from accessing their files or systems, commonly by encrypting data.
- How it works: It encrypts files and may display a ransom demand in exchange for restoring access.
- How it spreads: It can spread through phishing emails, malicious attachments, compromised accounts, malicious downloads, or exploited vulnerabilities.
- Target: Personal computers, organizational systems, servers, and important data.
- Impact: It can cause data loss, financial damage, downtime, and disruption of business operations.
- Example: An organization discovers that its important files have been encrypted and can no longer be accessed.
- Prevention: Maintain regular and protected backups, keep software updated, use security tools, enable MFA, and be cautious with links and attachments.